Q.No 5:

Problem Description:
One day Priya was jogging and realized that her life was boring. Everything was red, even the streets in the best park were red.

Therefore she decided to make streets a little bit brighter. She know that every street in the park is a segment laying on the X axis with coordinates Xl, Xr (Xl ≤ Xr). Streets may intersect or overlap.

She chooses any subset of streets and paints them in green. After that she wants to get one continuous green segment. As she really likes number L the length of this segment has to be equal to L.

Your task is to determine if it is possible to choose some subset of streets and paint them to get one green segment with the length equal to L?

Constraints:
1 ≤ sum of all N ≤ 2 * 10^3
1 ≤ L ≤ 10^6
1 ≤ Xl ≤ Xr ≤ 10^6
1 ≤ N ≤ 20, 1 ≤ Xl ≤ Xr ≤ 200, holds for test cases worth 10% of the problem's score.
1 ≤ N ≤ 100, 1 ≤ Xl ≤ Xr ≤ 200, holds for test cases worth 20% of the problem's score.

Input Format:

The first line contains one integer T - the number of test cases. Each test case starts with two integers N and L, denoting the number of streets and Priya's favorite number L. The next N lines contain two integers Xl, Xr, denoting the left and right borders of the street.

Output Format:
Print the output in a separate lines "Yes" if it is possible to paint some street and "No" otherwise.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2
5 3
1 2
2 3
3 4
1 5
2 6
2 3
1 2
2 6
```

EXPECTED OUTPUT

```
Yes
No
```

Test Case 2

INPUT (STDIN)

```
2
5 3
1 4
2 3
3 5
5 1
5 6
2 3
2 4
1 3
```

EXPECTED OUTPUT

```
Yes
Yes
```

Code:

```
#include <stdio.h>
#include <stdlib.h>
typedef long long ll;
struct Street
{
    ll left;
    ll right;
};
int compare(const void *a, const void *b)
{
    struct Street *x = (struct Street *)a;
    struct Street *y = (struct Street *)b;

    if (x->left < y->left)
        return -1;
    if (x->left > y->left)
        return 1;
    if (x->right < y->right)
        return -1;
    if (x->right > y->right)
```

```
        return 1;

    return 0;
}
int main()
{
    ll t;
    scanf("%lld", &t);

    int first = 1;

    while(t--)
    {
        ll n, L;
        scanf("%lld %lld", &n, &L);
        struct Street a[n];
        for(ll i=0;i<n;i++)
        {
            scanf("%lld %lld", &a[i].left, &a[i].right);
        }

        qsort(a, n, sizeof(struct Street), compare);

        ll found = 0;

        for(ll i=0;i<n;i++)
        {
            ll start = a[i].left;
            ll maxright = start + L;
            ll cur_right = a[i].right;

            if(cur_right > maxright)
                continue;

            for(ll j=0;j<n;j++)
            {
                if(a[j].left > cur_right)
                    break;

                if(a[j].left >= start &&
                   a[j].right <= maxright &&
                   a[j].right > cur_right)
                {
                    cur_right = a[j].right;
                }

                if(cur_right==maxright)
```

```
        {
          found = 1;
          break;
        }
      }

      if(found)
        break;
    }

    if(first)
    {
      printf("\n");
      first = 0;
    }
    printf("Output:\n");

    if(found)
      printf("Yes\n");
    else
      printf("No\n");
  }

  return 0;
}
```

Output:

Test_case 1:

```
/Desktop/c program/tce
2
5 3
1 2
2 3
3 4
1 5
2 6
2 3
1 2
2 6
Output:
Yes

Output:
No
```

Test_case 2:

```
2
5 3
1 4
2 3
3 5
5 1
5 6
2 3
2 4
1 3
Output:
Yes

Output:
Yes
aadityajaiswal@Aadi
```